Tension Forces in a Hanging Object

SYNOPSIS

An object held in mid-air by two strings is the simplest two-dimensional equilibrium there is, and it hides a genuinely surprising result. The strings do not share the load in any fixed way — how hard they pull depends entirely on the *angle* they make, and as they flatten toward horizontal the tension does not level off. It runs away to infinity. This note is where you resolve vectors into components and sum them separately, which is how every mechanics problem from here on is going to work.

BEFORE YOU READ ON — PREDICT

A heavy sign hangs from two wires fixed to a wall. It sags in the middle and looks untidy, so you tighten both wires to pull it up closer to horizontal.

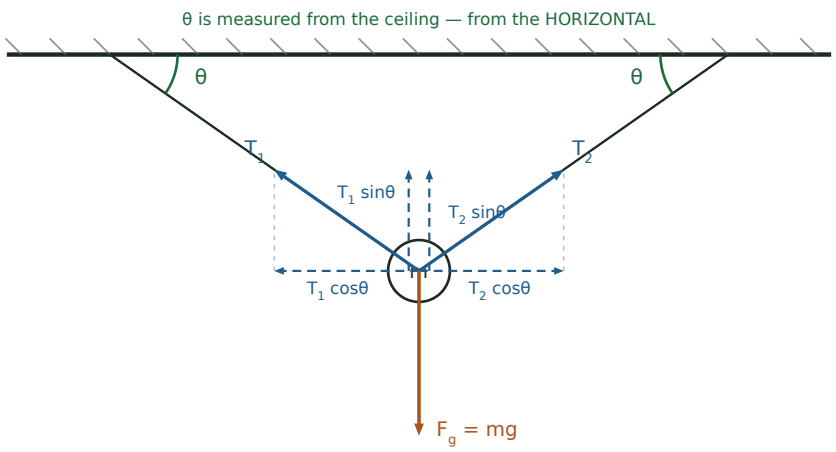
Does tightening the wires make them **safer**, or **more likely to snap**?

Commit to one, with a sentence of reasoning, before you turn the page.

PERFORMANCE EXPECTATIONS

1. **Draw** the free-body diagram of an object hung by two strings, and state which way a tension force acts.
2. **Resolve** each tension into horizontal and vertical components and build the component table, with the correct sign on each entry.
3. **Write** $\Sigma F_x = 0$ and $\Sigma F_y = 0$ as two independent equations, and use the first to show $T_1 = T_2$ when the angles are equal.
4. **Derive** $T = mg / (2 \sin \theta)$ and use it.
5. **Predict** how the tension changes as the strings are lengthened or shortened, and explain why a perfectly horizontal string is impossible.
6. **Explain** why the vertical components are the same at every angle while the horizontal components are not.

The setup



1. Draw this free-body diagram yourself. Two tension forces pointing away from the object along the strings, and the weight. Mark θ at each string.

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2. Resolve each tension into its two perpendicular components, head to tail. Which side of the right-angled triangle is adjacent to θ, and which is opposite?

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3. Write T_{1x}, T_{1y}, T_{2x} and T_{2y} in terms of T and θ. **Watch the signs** — a vector pointing left takes a negative sign, and so does one pointing down.

Two directions, two equations

The object is in equilibrium horizontally *and* vertically, so the forces add to zero in each direction **separately**. That is two independent equations, and it is the habit this note is really teaching. Fill the table from your own diagram.

	T_1	T_2	F_G	SUM
x-component			N/A	
y-component				

4. The strings are the same length, so the angles are the same. What does the **x-equation** then tell you about T_1 and T_2?

5. Use the **y-equation**, with $T_1 = T_2 = T$, to show that $T = mg / (2 \sin \theta)$.

The key relationships

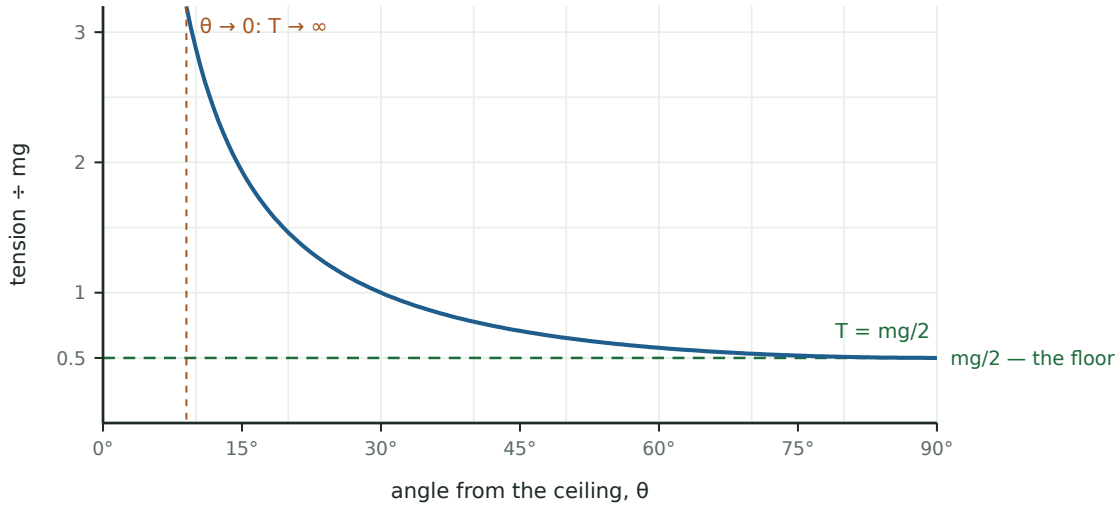
Eq. 1 $2T\sin\theta = mg$ — the vertical components carry the whole weight, between them

Eq. 2 $T = \frac{mg}{2\sin\theta}$ — **equal angles only.** θ from the horizontal

Eq. 3 $T_y = \frac{mg}{2}$ — each string's vertical pull — **the same at every angle**

Eq. 4 $T_x = \frac{mg}{2\tan\theta}$ — the horizontal pull — **unbounded** as θ → 0

What the angle does



6. An object of mass 2 kg hangs with the strings at 30° from the ceiling. Find the tension in each string.

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7. You replace the strings with **longer** ones, same anchor points. What happens to θ, and what happens to T? Explain using the equation.

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8. Now use strings so short that they are almost horizontal. What happens to T, and what would that mean in practice?

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9. If you want the tension as **small as possible**, should the strings be long or short? What is the smallest tension you could ever achieve?

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10. You can never actually reach a 0° angle — perfectly horizontal strings. Why not? What happens to the string before you get there?

Why the two components behave so differently

The *horizontal* components are a different story. They are not holding anything up — they are pulling against each other, and nothing caps how hard. As the strings flatten, each must pull harder and harder sideways just to keep its vertical share at mg/2. That is where the tension runs away, and it is why the answer to the sign question is **more likely to snap**.

11. As θ increases, what happens to the **vertical** component of the tension? And to the **horizontal** component? Justify both from the equations.

Check it on the bench

12. Set up a 2 kg object hung by two strings at **30° from the horizontal**. Calculate the tension first. Then measure it, and compare. If your measured value sits below the calculation, check your angle before you doubt the physics — a few degrees matters a great deal down at this end of the curve.

> **WITH CALCULUS**
>
> The bench instructions above warn that “a few degrees matters a great deal down at this end of the curve.” That is a claim about a *rate*, and a rate is something you can work out rather than take on trust.

$$T(\theta) = mg / (2 \sin \theta)$$

13. Differentiate with respect to θ. Write dT/dθ.

14. What is the **sign** of dT/dθ for every angle between 0° and 90°, and what does that sign alone tell you? Check it against your answer to task 8.

15. Evaluate dT/dθ for the 2 kg object at θ = 60° and again at θ = 5°. How many newtons does **one degree** cost at each end? (Your derivative is per radian — convert.)

16. For small θ, replace sin θ by θ. What simple form does T take? What happens to the tension if you then **halve** the angle?

17. Which factor in dT/dθ is responsible for the runaway — and why does the vertical component mg/2 not run away with it?

BACK TO YOUR PREDICTION

You predicted whether tightening the wires makes them safer or more likely to snap. Was it right? Point to the term in T = mg / (2 sin θ) that decides it.

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